

What to Evaluate?

- iterative design and evaluation is a continuous process that examines: early ideas for a conceptual model; early prototypes of the new system; later, more complete prototypes
- Designers need to check that they understand users' requirements
- Users want systems that are easy to learn and to use as well as effective, efficient, safe, and satisfying. Being entertaining, attractive, and challenging, etc. is also essential for some products.
- So, knowing what to evaluate, why it is important, and when to evaluate are key skills for interaction designers.

Why Evaluate?

- because user experience can be extremely important for a product's success
- - because the cycle of **design** and **testing** is the only validated methodology in existence that will consistently produce successful results
- ***Five good reasons for investing in user testing (Tognazzini):***
 - fixed problems before the product is shipped
 - the team can concentrate on real problems
 - Engineers code instead of debating
 - Reduced time to market
 - sell without bothering to release patches

When to Evaluate

- when it's brand new: develop markups of the product to elicit reactions from the potential users
- upgrades: compare user performance & attitude w/ previous versions

Evaluation Methods

- **Formative Evaluation:** *done at different stages of development to check that the product meets users needs*
- **Summative Evaluation:** *assess the quality of a finished product*
 - it is best to evaluate throughout design- from the first descriptions / sketches, all the way to the final product
 - Design proceeds through iterative cycles of design / test / re-design - where evaluation is a key ingredient for a successful design

An Evaluation Framework / Evaluation Paradigms

- User studies focus on how people behave, *in their natural environments*, or *in the laboratory*, with *old technologies* and with *new ones*.
- any kind of evaluation is guided by a set of beliefs, implicitly or explicitly.
- These beliefs are often supported by a theory. The beliefs and the methods associated with them are called ***evaluation paradigms***.
- There are four main evaluation paradigms discussed:
 - **Quick & Dirty:**
 - this is the common practice by which designers get feedback from users, to confirm that they are in line with users' needs and are liked. Emphasis is on fast input to the design process rather than carefully documented findings.
 - **Usability Testing:**
 - records typical users' performance on typical tasks in controlled settings. As the users perform the tasks they are watched and recorded on video, with keypresses logged. Data is used to calculate performance times, identify errors and help explain why the users did what they did.
 - User satisfaction questionnaires & interviews are used to gather users' opinions.

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An Evaluation Framework / Evaluation Paradigms (Contd.)

– **Field Studies:**

- done in natural settings- aims to understand what users do naturally and how technology impacts them. This can help to:
 - identify opportunities for new technology
 - determine design requirements
 - decide how best to introduce new technology
 - evaluate technology in use

– **Predictive Evaluation:**

- experts apply their knowledge of typical users to predict usability problems
- can involve heuristics, and theoretically based models
- users do not need to be present, and this method is cheap and quick to do
- GOMS / KLM / Fitt's Law???

Evaluation Techniques

- Observing users
 - can help identify needs, which can lead to new products
 - can help evaluate prototypes
 - ways in which to record observations: notes, audio, video, interaction logs
 - challenges: how can we observe others without disturbing them? how will we analyze the data gathered?
- Asking users their opinions
 - Techniques: Interviews and Questionnaires
 - Asking experts their opinions is inexpensive and quick
- Testing users' performance
 - ways to measure user performance to compare 2 or more designs
 - modeling users' task performance to predict the efficacy and problems of a user interface
 - some of these techniques: GOMS and the keystroke model

DECIDE: A framework to guide evaluation

- **Determine** the goals the evaluation addresses (what, who, why...)
- **Explore** the specific questions to be addressed (break down into sub questions- e.g. consumers' attitudes, security, interface, reputation of system, trust, adequate access)
- **Choose** the evaluation paradigm and techniques to answer the questions (paradigm selected determines the techniques used. Practical issues, ethical issues, and tradeoffs must be considered)
- **Identify** the practical issues (select users, stay on budget, stay on schedule, find evaluators, select equipment)
- **Decide** how to deal with the ethical issues (informed consent form, privacy / confidentiality, and let the participants know the goals of the study, what will happen to the findings, privacy of information, etc..)
- **Evaluate**, interpret, and present the data (what data to collect, how to analyze and present depends on the paradigm used. Need to consider reliability, validity, biases, scope and ecological validity)

Table 10 Summary of the usability methods covered in Chapters 5–7.

<i>Method Name</i>	<i>Lifecycle Stage</i>	<i>Users Needed</i>	<i>Main Advantage</i>	<i>Main Disadvantage</i>
Heuristic evaluation	Early design, "inner cycle" of iterative design	None	Finds individual usability problems. Can address expert user issues.	Does not involve real users, so does not find "surprises" relating to their needs.
Performance measures	Competitive analysis, final testing	At least 10	Hard numbers. Results easy to compare.	Does not find individual usability problems.
Thinking aloud	Iterative design, formative evaluation	3–5	Pinpoints user misconceptions. Cheap test.	Unnatural for users. Hard for expert users to verbalize.
Observation	Task analysis, follow-up studies	3 or more	Ecological validity; reveals users' real tasks. Suggests functions and features.	Appointments hard to set up. No experimenter control.
Questionnaires	Task analysis, follow-up studies	At least 30	Finds subjective user preferences. Easy to repeat.	Pilot work needed (to prevent misunderstandings).
Interviews	Task analysis	5	Flexible, in-depth attitude and experience probing.	Time consuming. Hard to analyze and compare.
Focus groups	Task analysis, user involvement	6–9 per group	Spontaneous reactions and group dynamics.	Hard to analyze. Low validity
Logging actual use	Final testing, follow-up studies	At least 20	Finds highly used (or unused) features. Can run continuously.	Analysis programs needed for huge mass of data. Violation of users' privacy.
User feedback	Follow-up studies	Hundreds	Tracks changes in user requirements and views.	Special organization needed to handle replies.

References

- <http://www.sharritt.com/CISHCIExam/preece.html>